THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The diagonal of a cuboid measuring 12 cm $\times$ 24 cm $\times$ 24 cm is: 1

(a) 36 cm **(b)** 35 cm **(c)** 37 cm **(d)** 38 cm

2. Take $\pi = \frac{22}{7}$. A toy is a cone of height 6 cm placed on a hemisphere, both of radius 21 cm. Its total volume is: 1

(a) 24394 cm^3 **(b)** 26613 cm^3 **(c)** 19958 cm^3 **(d)** 22176 cm^3

3. A solid sphere of radius 5 cm is melted and recast into small solid spheres of radius 1 cm. The number of small spheres is: 1

(a) 150 **(b)** 130 **(c)** 120 **(d)** 125

4. A cuboid 54 cm $\times$ 12 cm $\times$ 6 cm is melted and recast into cubes of edge 3 cm. The number of cubes is: 1

(a) 146 **(b)** 148 **(c)** 144 **(d)** 150

5. A cylinder of radius 7 cm and height 15 cm is melted and recast into cones of the same radius and height 1 cm. The number of cones is: 1

(a) 44 **(b)** 60 **(c)** 46 **(d)** 45

6. Take $\pi = \frac{22}{7}$. The capacity of a hemispherical bowl of inner radius 10.5 cm is: 1

(a) 2425.5 cm^3 **(b)** 2912.5 cm^3 **(c)** 2182.5 cm^3 **(d)** 2668.5 cm^3

7. Take $\pi = \frac{22}{7}$. The volume of a cone of radius 31.5 cm and height 42 cm is: 1

(a) 39293 cm^3 **(b)** 52392 cm^3 **(c)** 48025 cm^3 **(d)** 43659 cm^3

8. Take $\pi = \frac{22}{7}$. The volume of a cylinder of radius 35 cm and height 38 cm is: 1

(a) 175562 cm^3 **(b)** 146300 cm^3 **(c)** 131669 cm^3 **(d)** 160931 cm^3

9. **Assertion (A):** The volume of a cube of side 5 cm is 125 cm^3. 1

Reason (R): The volume of a cube of side a is a^3.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The total surface area of a cube of side 4 cm is 96 cm^2. 1

Reason (R): The lateral surface area of a cube of side a is $4a^2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A cuboid measures 6 cm × 18 cm × 27 cm. Find the length of its diagonal, its total surface area and its volume. **2**

12. Take $\pi = \frac{22}{7}$. A toy is a cone of radius 42 cm and height 56 cm resting on a hemisphere of the same radius. Find its total volume and its total surface area (curved parts only). **2**

13. Take $\pi = \frac{22}{7}$. A cylinder of height 4 cm has volume 5544 cm^3. Find its radius and its curved surface area. **2**

14. A cuboid measures 4 cm × 8 cm × 8 cm. Find the length of its diagonal, its total surface area and its volume. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A solid sphere of radius 12 cm is melted and recast into a solid cylinder of radius 3 cm. Find the height of the cylinder and its curved surface area (Take $\pi = \frac{22}{7}$). **3**

16. Take $\pi = \frac{22}{7}$. Water from a full cylindrical tank of radius 21 cm and height 15 cm is poured into an empty rectangular tank with a base 33 cm × 14 cm. Find the height of water in the rectangular tank. **3**

17. Take $\pi = \frac{22}{7}$. A hemispherical bowl has inner radius 21 cm. Find its capacity, and the number of such bowls that can be filled from a vessel holding 58212 cm^3 of soup. **3**

OR

A solid sphere of radius 6 cm is melted and recast into a solid cylinder of radius 3 cm. Find the height of the cylinder and its curved surface area (Take $\pi = \frac{22}{7}$). **3**

18. Take $\pi = \frac{22}{7}$. Water from a full cylindrical tank of radius 14 cm and height 20 cm is poured into an empty rectangular tank with a base 22 cm × 14 cm. Find the height of water in the rectangular tank. **3**

OR

Take $\pi = \frac{22}{7}$. A hemispherical bowl has inner radius 10.5 cm. Find its capacity, and the number of such bowls that can be filled from a vessel holding 14553 cm^3 of soup. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A school in Surat has a water tank in the shape of a cuboid, 5 m long, 4 m wide and 3 m deep, with an open top. **5**

(i) Find the volume of the tank.

(ii) Find the capacity of the tank in litres.

(iii) Find the total surface area of the closed cuboid of the same size.

(iv) Find the area of the inner surface of the open tank to be painted, and the cost at Rs 15 per m^2.

20. Take $\pi = \frac{22}{7}$. An ice-cream cone of radius 21 cm and height 28 cm is completely filled and topped with a hemisphere of ice-cream of the same radius. Find the total volume of ice-cream and the volume needed for 31 such cones. **5**

OR

A solid sphere of radius 6 cm is melted and 5 identical solid cylinders of radius 3 cm are made from the metal. Find the height of each cylinder. **5**

END OF QUESTION PAPER — Total: 40 Marks